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Antitumor activity of human γδ T cells transducted with CD8 and with T-cell receptors of tumor-specific cytotoxic T lymphocytes

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The difficulty in the induction and preparation of a large number of autologous tumor-specific cytotoxic T lymphocytes (CTL) from individual patients is one of major problems in their application to adoptive immunotherapy. The present study tried to establish the useful antitumor effectors by using γδ T cells through tumor-specific TCRαβ genes transduction, and evaluated the efficacy of their adoptive transfer in a non-obese diabetic/severe combined immunodeficiency (NOD/SCID) mice model. The TCRαβ gene was cloned from the HLA-B15-restricted CTL clone specific of the Kita-Kyushu Lung Cancer antigen-1 (KK-LC-1). The cloned TCRαβ as well as the CD8 gene were transduced into γδ T cells induced from peripheral blood lymphocytes (PBL). Cytotoxic T lymphocyte activity was examined using a standard 4 h ^{51}Cr release assay. Mice with a xenotransplanted tumor were treated with an injection of effector cells. Successful transduction of TCRαβ was confirmed by the staining of KK-LC-1-specific tetramers. The γδ T cells transduced with TCRαβ and CD8 showed CTL activity against the KK-LC-1-positive lung cancer cell line in a HLA B15-restricted manner. Adoptive transfer of the effector cells in a mice model resulted in marked growth suppression of KK-LC-1- and HLA-B15-positive xenotransplanted tumors. Co-transducing TCRαβ and CD8 into γδ T cells yielded the same antigen-specific activity as an original CTL *in vitro* and *in vivo*. The TCRαβ gene transduction into γδ T cells is a promising strategy for developing new adoptive immunotherapy. (*Cancer Sci* 2012; 103: 1414–1419)

Lung cancer is the most common malignant neoplasm and the leading cause of cancer mortality in industrialized countries.[1] In spite of advances in diagnostic and therapeutic approaches against lung cancer, there has been limited improvement in treatment outcome. Recent clinical studies on immunotherapy indicate favorable therapeutic effects, and might become one of the alternative treatment approaches for lung cancer.[2–4] The adoptive transfer of tumor-infiltrating lymphocytes after the administration of lymphodepleting preparative regimen mediates objective cancer regression in 50% of patients with metastatic melanoma.[5] However, the difficulty in the induction and expansion of a large number of autologous tumor-specific cytotoxic T lymphocytes (CTL) from individual patients is one of the major problems in their application to adoptive immunotherapy. To overcome this drawback, T-cell receptors (TCR) can be harnessed with antitumor specificities via molecular techniques. T-cell receptors with known antitumor reactivity can be genetically introduced into primary human T lymphocytes and provide effective tools for immunogenic therapy of tumors.[6]

In contrast, human γδ T cells can recognize and respond to a wide variety of stress-induced antigens, thereby developing innate broad antitumor and anti-infective activity.[7] These cells recognize antigens in a HLA complex-independent manner and develop strong cytolytic and Th1-like effector functions.[7] Furthermore, human γδ T cells can be activated by phospho-antigens and aminobisphosphonates. Aminobisphosphonates also facilitate large-scale *ex vivo* expansion of functional γδ T cells from the peripheral blood of cancer patients.[8] Although the antitumor effect is not antigen-specific cytotoxicity, γδ T cells are attractive candidate effector cells for cancer immunotherapy.

Kita-Kyushu Lung Cancer antigen-1 (KK-LC-1) is a recently identified cancer/testis (CT) antigen from lung adenocarcinoma.[9] KK-LC-1 is a cancer/testis antigen because it is not expressed in normal tissues except for the testis, and is located on the X chromosome (Xq 22). A 9-mer peptide (KK-LC-$1_{76\text{-}84}$; RQKRILVNL) is recognized by CTL in a HLA B15-restricted manner. The present study tried to establish a useful antitumor effector by using γδ T cells through tumor-specific TCR αβ genes transduction derived from a KK-LC-1-specific CTL clone, and evaluated the efficacy of their adoptive transfer in a severe combined immunodeficiency (SCID) mice model.

Materials and Methods

The study protocol was approved by the Human and Animal Ethics Review Committee of University of Occupational and Environmental Health and a signed consent form was obtained from each patient before taking the tissue samples used in the present study.

Culture medium. The culture medium for the cell lines was RPMI 1640 (GIBCO-BRL, Grand Island, NY, USA) supplemented with 10% heat-inactivated fetal calf serum (FCS; Equitech-Bio, Ingram, TX, USA), 10 mM HEPES, 100 U/mL of penicillin G and 100 mg/mL of streptomycin sulfate. The culture medium for γδ T cells and γδ T cells transduced with TCRαβ and CD8 genes was AlyS203-700 (Cell Science and Technology Institute, Sendai, Japan) supplemented with 10% heat-inactivated FCS. The γδ T cells were expanded with 1 μM Zoledronate (Novartis, Basel, Switzerland) followed by the addition of 100 units/mL interleukin-2 (IL-2) (Fig. 1).

Cell lines. F1121L, A110L and B901L lung adenocarcinoma cell lines were established from surgical specimens, which had the genotype of HLA-A*2402/0201, B*4006/1507, Cw*0303/0801, HLA-A*2402/, B*5201/, Cw*1201/, and HLA-A*0206/2601, B*3901/4006, Cw*0702/0801, respectively. The three lung adenocarcinoma cell lines showed a positive expression for KK-LC-1. The method used to establish the lung cancer cell line has been described previously.[10] F1121 Epstein–Barr virus-transformed B cells (F1121 EBV-B) were derived from

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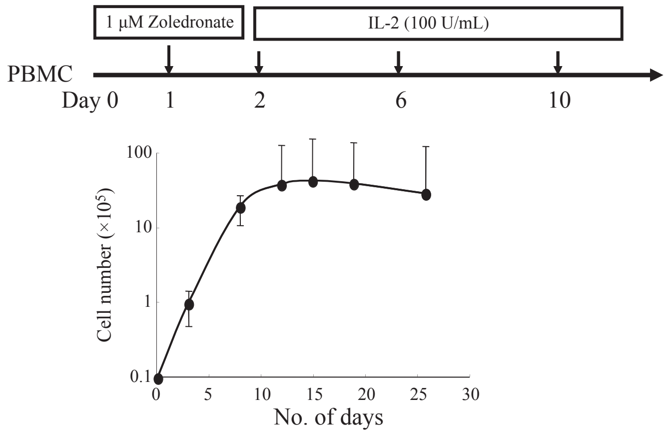

Fig. 1. Proliferation of γδ T cells in the presence of zoledronate and interleukin-2 (IL-2). The γδ T cells expanded 400-fold 2 weeks after stimulation with zoledronate in the presence of IL-2 100 U/mL.

peripheral blood lymphocytes (PBL) of F1121 treated with 1 μg/mL cyclosporine A (Sandoz, Basel, Switzerland) and 20% of the supernatant of the Epstein–Barr virus-transformed marmoset monkey lymphocyte B95-8. K562 is an erythroleukemia cell line that is sensitive to natural killer cell cytotoxicity. Tumor necrosis factor (TNF)-sensitive WEHI 164c13 cells were kindly donated by Dr Coulie PG (Cellular Genetics Unit, Universite Catholic de Louvaine, Brussels, Belgium). WEHI-164c13 cells were maintained in culture medium with 5% FCS.

TCRαβ gene cloning from KK-LC-1$_{76\text{-}84}$-specific CTL clone. The TCRαβ gene was cloned from the HLA-B15-restricted CTL clone specific for KK-LC-1 (cancer/testis antigen), which was identified from a patient with lung adenocarcinoma.[9] The cloned TCRα and TCRβ were then joined by a picornavirus-like 2A 'self-cleaving' peptide by overlapping PCR. The short 18 amino acid 2A sequence that separates the TCR α and TCR β results in equimolar expression of the TCRαand TCRβvia a 'ribosomal skip' mechanism.[11] The TCR α and TCR β combined with the 2A sequence were cloned into a pMX retroviral vector. The pMX vector harbors a 5′ long terminal repeat (LTR) and the extended packaging signal derived from a MFG vector followed by a multi-cloning site suitable for cDNA construction and 3′ LTR of Moloney murine leukemia virus. The resulting vector pMX in combination with Plat-A cells (Cell Biolabs Inc., San Diego, CA, USA) produces an average of 1×10^7 IU/mL of virus.[12] pMX-vector had a conjugated puromycine-resistant gene for the selection of the transductant.

TCRαβ gene transduction using retroviral vector. An infectious, replication-incompetent retrovirus was produced using the pMX retroviral vector system with Plat-A cells and the Lipofectamine 2000 reagent (Invitrogen, Carlsbad, CA, USA). Briefly, Plat-A cells were transfected with the TCR containing plasmid. After 1 day culture from the transfection, supernatant of Plat-A cells was harvested, filtered using a 0.22 mm sterile filter and concentrated by centrifugation at 5800*g* overnight. Concentrated supernatant containing retrovirus was used for the gene transduction. The cloned TCR αβ as well as the CD8 gene (provided by Takara Bio Inc., Otsu, Japan) were transduced into γδ T cells induced from PBL by using a retroviral vector in recombinant human fibronectin fragment CH-296 (Retronectin, Takara Bio)-coated six-well plates (Nunc, Roskilde, Denmark). The transduction was repeated the following day. Cytotoxic T lymphocyte activity was examined using a cytotoxicity assay and a cytokine production assay.

KK-LC-1/HLA-B15 tetramer staining. The transduction of TCRαβ was confirmed by the staining of KK-LC-1-specific tetramers. KK-LC-1-specific tetramers (T-Select MHC Tetramer) were purchased from Medical & Biological Laboratories Co., Ltd (Nagoya, Japan). TCRαβ-transduced γδ T cells were washed and resuspended in PBS with 1% human AB serum, and incubated for 30 min at 37°C with the KK-LC-1$_{76\text{-}84}$/HLA-B15 tetramer (20 nM each) coupled with phycoerythrin. The cells were washed, fixed with 0.5% formaldehyde and analyzed on a FACS Calibur flow cytometer (BD Biosciences, San Jose, CA, USA) using the FlowJo software package (Tree Star Inc., OR, USA).

Monoclonal antibody (mAb) for cytotoxicity assay and cytokine production. Hybridomas (HB-145, HB-95) were purchased from the American Type Culture Collection (ATCC, Rockville, MD, USA). C7709.A2.6 (anti-HLA-A24) and B1.23.2 (anti-HLA-B, C) were kindly donated by Dr Coulie PG. The culture supernatants of ATCC HB–145 (IVA12; anti-HLA-DR, DP, DQ) and HB-95 (W6/32; anti-HLA-A, B, C) were used for analyzing the HLA restriction of CTL and antitumor effectors. The anti-NKG2D antibody was purchased from BD Biosciences.

Cytotoxicity assay and cytokine production of CTL. The cytotoxicity of CTL was assessed using a standard 4 h ^{51}Cr release assay as described previously.[13] The TNF production of CTL was measured using a WEHI assay using TNF-sensitive WEHI cells.[13] Briefly, CTL clone L7/8 (6×10^4/mL) TCR-transduced γδ T cells was incubated with tumor cells (6×10^5/mL) in culture medium with 10% FCS overnight and the amount of TNF in the culture supernatant was measured in a triplicate assay by evaluation of the cytotoxic effect on WEHI-164c13 cells in a MTT colorimetric assay. Supernatants were also collected to measure interferon–γ (IFN-γ) production in a triplicate assay using an IFN-γ ELISA test kit (Life technologies, Inc., Gaithersburg, MD, USA) according to the manufacturer's instructions. In the blocking assay using mAb, a¼-diluted culture supernatant of hybridomas such as HB-95, C7709.A2.6, B1.23.2 and HB-145 was used for the antibody inhibition assay.

Lung adenocarcinoma xenograft model. The γδ T cells were expanded from peripheral blood mononuclear cells (PBMC) of patients with adenocarcinoma with 100 units/mL rIL-2 after stimulation with zoledronate. The number of γδ T cells was calculated with a flow cytometer by using anti-TCR γδ ((BD Biosciences). The activated γδ T cells were transduced with TCR αβ gene derived from a KK-LC-1-specific CTL clone; the antitumor effect was assessed in a lung adenocarcinoma (B901L) xenotransplanted non-obese diabetic/severe combined immunodeficiency (NOD/SCID) mouse model. The parental B901L cell line expresses KK-LC-1 but does not possess the HLA-B15 molecule. B901L-parental and HLA-B15 transduced B901L were inoculated subcutaneously with 1×10^6 cells in the lateral flank of a NOD/SCID mouse at day 0. TCRαβ and CD8 transduced γδ T cells were injected via the tail vein of immunodeficient mice (NOD/SCID mice) weekly or twice weekly. Vehicle (PBS) was injected intravenously in the same manner. The effects of treatment were evaluated by measuring tumor size. The volume of the tumor was calculated using the formula: $v = 0.4 \times a \times b^2$, where a is the maximum diameter of the tumor, and b is the diameter at a right angle to a.

Results

***In vitro* expansion of γδ T cells.** The γδ T cells could easily be expanded with 1 μM of zoledronate in the presence of IL-2 100 U/mL. Flow cytometry of the cultured cells revealed that the population of γδ T cells in the PBMC was 2–3% initially, and increased more than 95% at day 14. The number of γδ T cells could be expanded approximately 400-fold during the 2 weeks after stimulation with zoledronate (Fig. 1). The growth curves of the γδ T cells are shown as mean ± standard deviation on the basis of three independent experiments.

Transduction of the TCRαβ gene of tumor-specific CTL. The TCRαβ gene was cloned from the HLA-B15-restricted CTL clone specific for KK-LC-1 (cancer/testis antigen), which was established from a patient with lung adenocarcinoma. The original CTL clone specific for KK-LC-1 showed positive staining with the KK-LC-$1_{76\text{-}84}$/HLA-B15 tetramer. The γδ T cells were negative for the KK-LC-$1_{76\text{-}84}$/HLA-B15 tetramer. The cloned TCRαβ and CD8 genes were co-transduced into the γδ T cells. After transduction of the TCRαβ gene, expression of the TCR specific for KK-LC-1 was confirmed by flow cytometry staining with the KK-LC-$1_{76\text{-}84}$/HLA-B15 tetramer (Fig. 2). The tetramer staining was positive more than 80% after puromycine selection. The data of flow cytometry shown are representative of at least three independent experiments. The effector cells were used for further experiments when the staining for the KK-LC-$1_{76\text{-}84}$/HLA-B15 tetramer was confirmed to be more than 80% positive.

Cytotoxic T lymphocyte activity of γδ T cells co-transducted with the TCR gene and the CD8 gene. The CTL activity against the KK-LC-1 epitope was examined using a cytotoxicity assay and cytokine production assay. The γδ T cells indicated strong natural killer (NK) activity and non-specific cytotoxicity against the allogeneic lung cancer cell line (Fig. 3A). The γδ T cells transduced with only the TCR gene showed neither CTL activity nor NK activity (Fig. 3B). NKG2D expression was examined to evaluate inhibition of the cytotoxicity of the γδ T cells after transduction with the TCR gene. The expression of NKG2D was remarkably suppressed after TCR transduction (Fig. 4). In contrast, the γδ T cells co-transfected with the TCR and CD8 genes (TCRαβ-CD8 γδ T cells) showed tumor antigen-specific cytotoxic activity that was similar to that of the original CTL clones (Fig. 3C). The NK cytotoxic activity and non-specific cytotoxicity were dramatically inhibited after TCR transduction (Fig. 3C). TCRαβ-CD8 γδ T cells produced IFN-γ in response to F1121L adenocarcinoma as well as EBV-B cells pulsed with KK-LC-1 peptide (Fig. 5). The production was blocked by anti-HLA class I antibody, anti-HLA B/C antibody and anti-CD8 antibody.

Antitumor activity of TCRαβ-CD8 γδ T cells *in vivo*. For evaluation of the effect of TCRαβ-CD8 γδ T cells *in vivo*, a HLA-B15-positive B901L (lung adenocarcinoma cell line) was established by stable transfection of the HLA-B15 gene. Parental B901L is negative for HLA-B15 but positive for KK-LC-1, and therefore the TCRαβ-CD8 γδ T cells showed cytotoxic activity against HLA-B15 transfected B901(B901L-HLA-B15), but not against parental B901L (Fig. 6). Intravenous injection of TCRαβ-CD8 γδ T cells inhibited growth of the susceptible cancer cell line of B901L-HLA-B15. B901L-parental and B901L-HLA-B15 were xenotransplanted subcutaneously in the lateral flank of a NOD/SCID mouse on day 0. The growth of the susceptible cancer cell line (B901L-HLA-B15) was inhibited significantly by weekly intravenous injection of the TCRαβ-CD8 γδ T cells (1×10^6/injection) in comparison to the control cell line (B901L; Fig. 7A). The growth rate of both cell lines was almost the same without the adoptive immunotherapy (data not shown). The growth of B901L-HLA-B15 was remarkably decreased by twice-weekly injection of the TCRαβ-CD8 γδ T cells (Fig. 7B).

Infiltration of adoptively transferred TCRαβ-CD8-γδ T cells into the tumor. The histological examination and immunohistochemical staining revealed that the CD3-positive human lymphocytes had infiltrated into the susceptible tumor (B901L-HLA-B15). The infiltration of CD3-positive human lymphocytes was observed more strongly in the B901L-HLA-B15 tumor compared with the control cell line (B901L) (Fig. 8A). The central part of the tumor (B901L-HLA-B15) exhibited necrosis. Expression of the TCR Vα13 gene of the effector cells (TCRαβ-CD8-γδ T) was identified in the central part of the tumor tissue (B901L-HLA-B15) using RT-PCR. The original CTL specific for KK-LC-1 also possessed TCR Vα13 (Fig. 8B).

Discussion

The mechanism of the antitumor immune response has been gradually elucidated based on the recent progress of molecular biological technique. However, clinical studies have not shown a satisfactory clinical response rate. The clinical outcome is still inferior to established treatment such as chemotherapy and radiotherapy.[14] Therefore, immunotherapy must overcome several problems before it becomes an established therapy for cancer patients. Adoptive immunotherapy requires the induction of tumor-reactive T lymphocytes *ex vivo* followed by expansion of these cells in order to generate sufficient numbers for infusion. Although several studies demonstrate that autologous *ex vivo*-generated antitumor CTL can be administered safely in patients with advanced solid tumors and can improve the immunological antitumor reactivity in recipients, the strategy is hampered by the difficulty of isolating and expanding a sufficient quantity of autologous tumor-reactive T cells capable of preserving the cytotoxic capacity.[15,16]

Several investigators have reported TCR transduction technology with retroviral vectors enables generation of novel effectors that demonstrate HLA-restricted, antigen-specific CTL functions.[17,18] The transduction of the TCRαβ genes facilitates production of a large number of antitumor effectors functionally similar to CTL activity in a reproducible fashion without laborious techniques.[19] However, the disadvantage of TCRαβ transfer to other αβ T cells is the possible formation

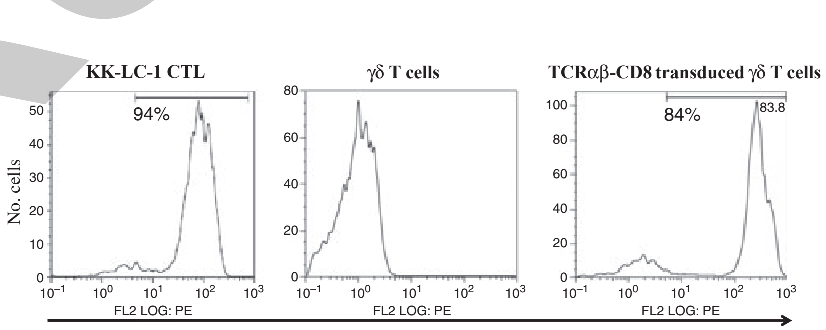

Fig. 2. KK-LC-$1_{76\text{-}84}$/HLA-B15 tetramer staining of γδ T cells transfected with TCRαβ and CD8 genes. Expression of TCRαβ was confirmed by the staining of HLA tetramers. The positive tetramer staining for specific KK-LC-$1_{76\text{-}84}$ reached more than 80% after puromycine selection.

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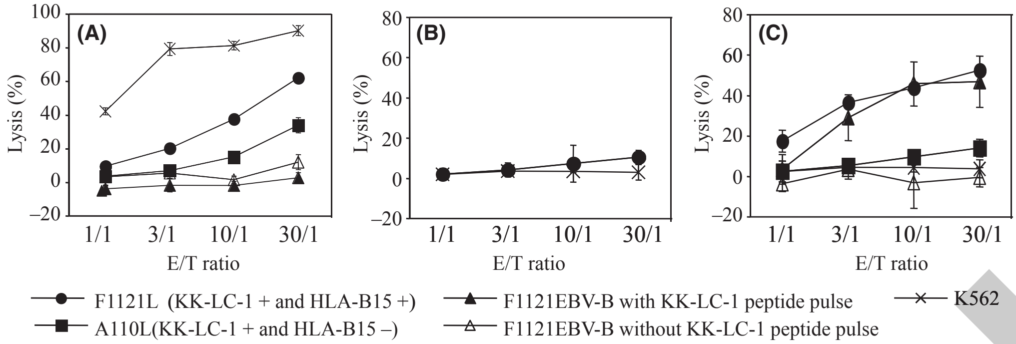

Fig. 3. Cytotoxic activity of γδ T cells transduced with TCRαβ gene derived from KK-LC-1-specific cytotoxic T lymphocytes (CTL). The γδ T cells co-transduced with TCRαβ and CD8 showed cytotoxic activity to KK-LC-1-positive tumor cells (F1121L) as well as F1121EBV-B pulsed with KK-LC-$1_{76\text{-}84}$. However the γδ T cells transduced with only *TCRαβ gene showed neither CTL activity nor natural killer (NK) activity. The effector cells were γδ T cells in (A), γδT cells transduced with only the TCR αβ gene in (B), and γδ T cells co-transduced with TCRαβ and CD8 genes in (C). E/T, effector/target.

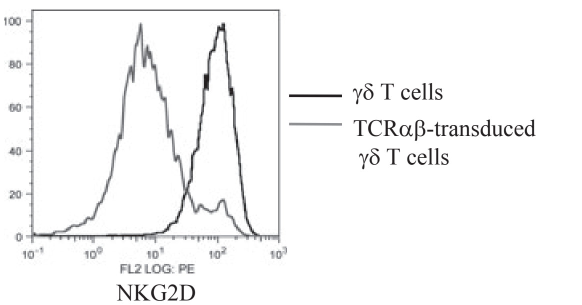

Fig. 4. Downregulation of NKG2D expression by transfection with the TCRαβ gene. NKG2D expression was measured using flow cytometry. The expression of NKG2D was remarkably suppressed after transduction of the TCR gene.

of mixed TCR heterodimers. The introduced α or β TCR could pair with the endogenous α or β TCR chains to generate unfavorable T cells with self-antigen specificity.[20,21] Okamoto *et al.*[22] reported the effect of small interfering RNA (siRNA) constructs that specifically downregulate endogenous TCR to inhibit formation of mispairing TCR heterodimers. The human lymphocytes transduced with siRNA exhibit high surface expression of the introduced tumor-specific TCR and reduced expression of endogenous TCR, and these lymphocytes transduced with tumor-specific TCR demonstrate enhanced cytotoxic activity against antigen-expressing tumor cells.

The γδ T cells account for 2–10% of T lymphocytes in human blood and also play a role in immune surveillance against microbial pathogens and cancer.[23] The γδ T cells recognize small non-peptidic phosphorylated compounds, referred to as phosphoantigens, via polymorphic γδ TCR, as well as the major histocompatibility complex class I chain-related molecules, A and B (MICA and MICB), via NKG2D receptors in a HLA-unrestricted manner.[24] The γδ T cell activation is induced by zoledronic acid through accumulation of isopentenyl diphosphate.[25] Recent studies have reported that aminobisphosphonates, currently used in cancer treatment for bone metastases, can activate γδ T cells both *in vitro* and *in vivo*.[26,27] These data suggest that γδ T cells could be an alternative population for efficient TCR transfer. Hiasa *et al.*[28] proposed γδ T cells as a target for retroviral transfer of cancer-specific TCR. They indicated that γδ T cells co-transduced with TCRαβ and CD8 alphabeta genes acquire cytotoxicity against tumor cells and produce cytokines in both αβ- and γδ-TCR- dependent manners.

The present study demonstrated that γδ T cells could be reproducibly proliferated from peripheral blood lymphocytes

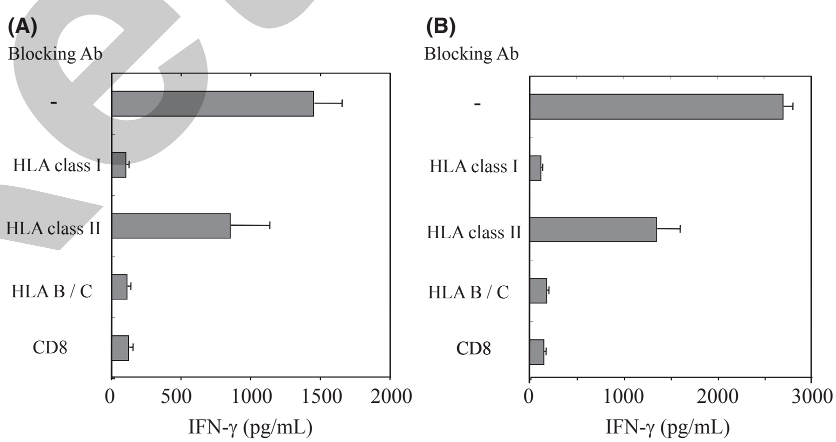

Fig. 5. Interferon-γ (IFN-γ) production of T-cell receptors (TCR) αβ-CD8 γδ T cells in response to the F1121L adenocarcinoma cell line and Epstein-Barr virus-B (EBV-B) cells pulsed with the KK-LC-1 peptide. The TCR αβ-CD8 γδT showed IFN-γ production in response to KK-LC-1-positive tumor cells as well as F1121EBV-B pulsed with KK-LC-$1_{76\text{-}84}$. The IFN-γ production was inhibited by the addition of an anti-HLA class I antibody, anti-HLA B/C antibody and anti-CD8 antibody. The stimulator was F1121L in (A) and F1121EBV-B pulsed with KK-LC-$1_{76\text{-}84}$ in (B). Results are expressed as mean ± SD.

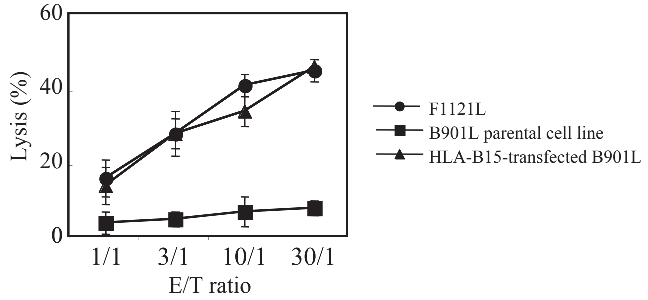

Fig. 6. Recognition of KK-LC-1-specific T-cell receptors (TCR) αβ-CD8-γδ T cells against HLA-B15-transfected B901L. A HLA-B15-positive B901L (lung adenocarcinoma cell line) was established by stable transfection of the HLA-B15 gene. The B901L-HLA-B15 cell line showed a similar susceptibility to F1121L as target cells. However, the TCRαβ-CD8 γδ T cells showed no cytotoxic activity against parental B901L. E/T, effector/target.

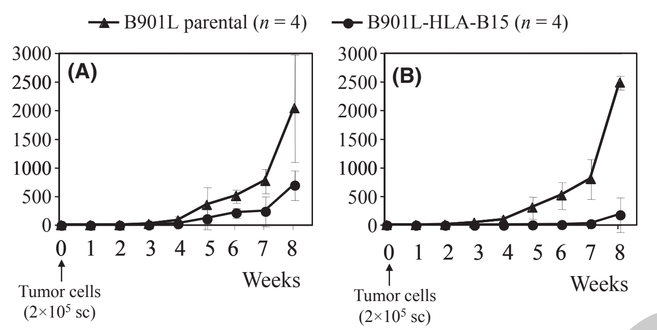

Fig. 7. Growth inhibition of B901L-HLA-B15 tumor by adoptive transfer of TCR αβ-CD8 γδ T cells. (A) Weekly intravenous injection of T-cell receptor (TCR) αβ-CD8 γδ T cells inhibited growth of the susceptible cancer cell line (B901L-HLA-B15) *in vivo*, but not for the parental cell line. (B) The twice-weekly injection of TCR αβ-CD8 γδ T cells remarkably inhibited the growth rate compared with the control cell line (parental B901L cell line). Results are expressed as mean tumor volume ± SD. sc, subcutaneous injection.

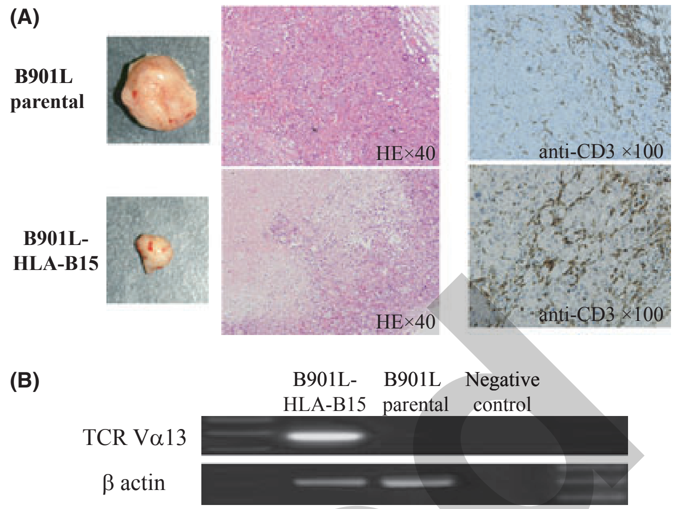

Fig. 8. Infiltration of adoptive transfer of T-cell receptor (TCR) αβ-CD8 γδT cells into the tumor. (A) Immunohistochemical staining with anti-human CD3 antibody. A large amount of CD3-positive human lymphocytes infiltrated into the B901L-HLA-B15 tumor. (B) Detection of TCR-Vα13 of TCRαβ-CD8-γδ T cells in the tumor using RT-PCR.

by stimulation with zoledronate. The effector cells (TCRαβ-CD8 γδ T cells) were then established from the proliferated γδ T cells by co-transduction of the TCRαβ and CD8 genes. The antigen specificity of the γδ T cells was confirmed using KK-LC-$1_{76\text{-}84}$/HLA-B15 tetramer staining. The effector cells showed tumor antigen-specific cytotoxic activity and cytokine production similar to those of the original CTL clones. However, γδ T cells transduced with TCRαβ without CD8 lost their non-specific cytotoxic activity. One of the possible causes of the decline in non-specific cytotoxic activity following the transfection of the TCRαβ genes was downregulation of the surface expression of NKG2D on the γδ T cells, because NKG2D is an important co-stimulatory molecule for γδ T cells. Intravenous injection of TCRαβ-CD8 γδ T cells inhibited growth of susceptible cancer cells (B901L-HLA-B15) *in vivo*. A histological examination revealed infiltration of TCRαβ-CD8 γδ T cells into the central part of the tumor, and the tumor subsequently became extensively necrotic. The gene transduction of TCRαβ and CD8 into γδ T cells is a promising strategy to develop a new adoptive immunotherapy against malignancies.

Cancer/testis antigens are particularly attractive targets for immunotherapy, because of their unique expression profiles that have restricted expression in testis and placental trophoblasts and male germ-line cells, which do not express HLA class I molecules and therefore cannot present the antigens to T lymphocytes.[29] Cancer/testis antigens are activated in a number of tumors of various histological types. KK-LC-1 is a cancer/testis antigen recognized by CTL. The expression rate of KK-LC-1 in patients with NSCLC is 32.6%, and the expression of KK-LC-1 is higher (32.1%) in patients with adenocarcinoma compared with other CT antigens (MAGE-A3, MAGE-A4 and NY-ESO-1).[30] KK-LC-1 might be a hopeful target for patients with adenocarcinoma, because in Japan the recent incidence of adenocarcinoma is twice that of squamous cell carcinoma.[31]

The reproducible effects of transduction of the TCRαβ gene by retroviral vector was confirmed using KK-LC-$1_{76\text{-}84}$/HLA-B15 tetramer staining. The co-transduction of TCRαβ and CD8 into these γδ T cells showed the same antigen-specific activity as an original CTL *in vitro* and *in vivo*. The gene transduction of TCRαβ and CD8 into γδ T cells is a promising strategy for developing a new adoptive immunotherapy against malignancies.

Acknowledgments

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Disclosure Statement

The authors have no conflict of interest.

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